

GitHub Waypoint

1. What the project is

Waypoint is a free web app (live at github-waypoint.vercel.app) that traces public GitHub activity. You type a **username**, an **organization**, or an **owner/repo**, and it shows where that account actually spends its time: lifetime pull requests and merge rate, the repositories they contribute to, review speed, label distribution, top contributors, likely maintainers, and recent PRs/issues. Repo results can be exported to CSV or JSON. There is no login and no backend database: all data comes from the public GitHub REST API, optionally using a personal access token that stays in the visitor's browser.

A companion page (/cli) markets a separate command-line tool (waypoint-cli) that performs the same traces in a terminal.

2. Technology stack

Technology	Version	Role in the project
Next.js (App Router)	16.2.10	Framework: file-based routing, server components, metadata API, ISR (revalidate), next/font, next/image, next/og, next/script.
React / React DOM	19.2.4	UI library. Hooks used: useState, useEffect, useLayoutEffect, useRef. react-dom createPortal for the modal.
JavaScript (ES2022+)	-	Whole codebase is plain JS/JSX (no TypeScript). jsconfig.json defines the @/* path alias.
Tailwind CSS	4.3.2	Utility classes (via @tailwindcss/postcss) alongside a large hand-written globals.css and a CSS module (page.module.css).
PostCSS	-	Build step that runs the Tailwind v4 plugin (postcss.config.mjs).
motion (Framer Motion)	12.42.2	Used in Reveal.jsx for fade-and-lift entrance animations.
@vercel/analytics	2.0.1	Page-view analytics component in the root layout.
ESLint + eslint-config-next	9 / 16.2.10	Linting (npm run lint).
GitHub REST API	v3 JSON	Only data source: /users, /orgs, /repos, /pulls, /issues, /search/issues, /events, /contributors.
Vercel	-	Hosting and deployment; ISR caching of server-fetched pages.
Local fonts	woff2	Matter (variable sans) and Matter SemiMono, loaded with next/font/local and preloaded.

3. Folder structure

Path	Contents
src/app/	Routes. page.jsx (home), [owner]/page.jsx (user or org), [owner]/[repo]/page.jsx (repo), cli/page.jsx, not-found.jsx, layout.jsx (root shell, fonts, SEO, theme script), plus metadata routes: sitemap.js, robots.js, manifest.js, opengraph-image.jsx.
src/lib/analyze.js	Client-side data engine: cached GitHub fetch wrapper and analyzeQuery(), which decides user vs org vs repo and gathers all data with parallel requests and pagination.
src/lib/analyzeServer.js	ghFetchLight(): one lightweight server-side fetch (5-minute revalidate) for page metadata and an instant header.
src/lib/utils.js	Pure helpers: buildTracePath, timeAgo, EVENT_LABELS, eventToTimelineItem, buildUserAnalysis, filterPRs, bucketPRsByTime.

Path	Contents
src/lib/services.js	CacheService (TTL cache + eviction in localStorage), HistoryService (recent searches), ExportService (JSON/CSV download).
src/components/	Result views (UserResult, OrgResult, RepoResult), LiveResult (loading/error/retry), ResultsClient, ResultLayout (shared bento/row UI), SearchBar, TokenBox, RecentSearches, ContributionsModal, AsciiDissolve, Reveal, NavBar, SiteFooter, Wordmark, AppWindow.
src/components/home/, cli/	Marketing sections for the home and CLI pages (Hero, HeroGrid, Features, Faq, Splits, TerminalShot, InstallCommand, ...).
src/content/	Static copy and data (home.js, cli.js, hero-grid.js coordinates) kept separate from components.
public/	Fonts, favicon and images.

4. How a trace works (request flow)

- **Search.** SearchBar trims input, strips a leading @ and trailing slashes, rejects more than two path segments, and navigates to a clean URL built by buildTracePath() (e.g. /vercel/next.js).
- **Server render.** The dynamic route runs on the server, calls ghFetchLight() for the user/repo object (cached for 300 s, optionally with a server GITHUB_TOKEN), produces SEO metadata with generateMetadata(), and calls notFound() on 404.
- **Client hydration.** LiveResult paints a light header immediately, then calls analyzeQuery() in the browser with the visitor's own token, so GitHub's rate limit is spread across visitors instead of shared on the server.
- **Data gathering.** analyzeQuery() fires independent requests together with Promise.all, paginates PRs up to 5 pages (500 items), and caches every response in localStorage for 10 minutes.
- **Analysis & render.** Results go to UserResult / OrgResult / RepoResult, which count, filter, sort and rank the data. The search is saved to history and the result appears with the ASCII dissolve animation.
- **Errors.** HTTP status codes become NOT_FOUND / RATE_LIMIT / HTTP_xxx errors. On rate limit the UI offers a token input and a Retry button.

5. Data structures used

All of them are JavaScript built-ins. No custom tree, graph, heap, linked list or trie is implemented.

Data structure	Where	Purpose
Hash map (plain Object)	utils.js buildUserAnalysis (repoStats, typeCounts); analyze.js contribMap; RepoResult labelCounts, contribCounts; UserResult repoPRs, repoIssues; EVENT_LABELS lookup table	Counting, grouping by key, and O(1) lookup.
Map (ordered hash map)	utils.js bucketPRsByTime (bucketsMap)	Time buckets in chronological insertion order, with O(1) key lookup.
Set	RepoResult (unique branches); analyze.js (unique member logins); UserResult (seen URLs)	De-duplication in O(n).
Array / list	Everywhere (PR, issue, event, timeline lists)	Ordered collections, processed with map / filter / reduce / sort / slice / concat.
Array used as MRU list (deque-like)	services.js HistoryService	Recent searches: unshift puts the newest at the front, capped at 20 entries.
Typed array (Float32Array)	AsciiDissolve.jsx	2D grid of random thresholds stored flat; index = $r * \text{cols} + c$ (row-major order).
Key-value store with expiry	services.js CacheService on localStorage	Cache of API responses; each entry is {data, expiry}.
Tuples / entry pairs	Object.entries(...) results	[key, value] pairs, sorted to produce rankings.
Coordinate records	content/hero-grid.js	Lists of {col, row, w, h} objects for the decorative hero grid.
2D table (rows x cells)	ExportService.exportToCSV	Array of rows serialized into CSV text.

6. Algorithms & techniques

Technique	Where	How it works / complexity
Frequency counting (hashing)	buildUserAnalysis, contributor, label and PR-per-repo counts	Single pass with counter[key]++; O(n).
Group-by	UserResult repoPRs / repoIssues	Buckets items into per-repo arrays keyed by name parsed from repository_url; O(n).
Comparison sort + Top-K	topRepos (8), topTypes (7), keyContributors (9), topLabels (8), top contributors (9)	Array.sort with a comparator, then slice(0, K); O(n log n). A min-heap would give O(n log k).
Multi-key, direction-aware sort	RepoResult sortedContributors	User picks the key (count, merged, open, closed, derived mergeRate) and asc/desc; the comparator flips its sign.
Filtering (linear scan)	filterPRs(days, branch); merged / open / closed splits	Cutoff = now - days * 86 400 000 ms; O(n).
Aggregation / reduce	Average days to merge	reduce sums (merged_at - created_at); divides by count.
Ratio metrics	Merge rate	merged / (merged + closed-unmerged) for repos; merged / total for users.
Merge + sort + de-duplicate	UserResult timeline	Concatenates events and PRs, sorts by date descending, skips repeated URLs with a Set, keeps 12.
Time bucketing (histogram)	bucketPRsByTime	Picks week or month granularity (range > 200 days means months), pre-fills empty buckets, snaps dates to Monday, counts; O(n + buckets).

Technique	Where	How it works / complexity
Cache with TTL	CacheService.get / set	Expired entries are deleted when read (lazy expiry).
Eviction policy	CacheService on QuotaExceededError	Collects cached GitHub keys, sorts by expiry ascending, deletes the oldest 50% (corrupt entries first), retries the write. Similar to FIFO/LRU eviction.
MRU list maintenance	HistoryService.addHistory	filter out the old copy, unshift the new one, truncate to 20; O(n).
Pagination + early termination	analyze.js (repo PRs, user PR search)	Fetches pages in parallel; stops at an empty/short page or a PR older than 6 months; page count = min(5, ceil(total / 100)).
Concurrency (fork-join)	Promise.all throughout analyze.js	Independent requests run in parallel, which cuts round trips.
Memoization of network calls	ghFetch keyed by URL	Repeated identical URLs are served from the cache.
String parsing / normalization	buildTracePath, SearchBar, repository_url.split('/')	Regex trimming, splitting, encodeURIComponent, validation of the segment count.
Escaping / serialization	exportToCSV, JSON.stringify	RFC-4180 style quoting (" doubled); Blob plus object URL download.
Random thresholds + easing	AsciiDissolve	Each grid cell gets a random threshold; easeOutQuart 1 - (1 - t)^4 drives progress; cells step down a character ramp.
Row-major indexing	AsciiDissolve	Nested row/column loop over a flat array; O(rows x cols) per frame.
Unit conversion arithmetic	timeAgo	ms -> minutes -> hours -> days -> months with integer division.

7. Framework features, web APIs & design patterns

Next.js / React

- Server Components (route pages) and Client Components ('use client') split by responsibility.
- Dynamic routes [owner] and [owner]/[repo]; async params; generateMetadata() for per-page SEO.
- Incremental Static Regeneration: export const revalidate = 300 and fetch(..., { next: { revalidate } })).
- notFound() and a custom not-found.jsx; next/navigation useRouter for client navigation.
- Metadata routes: sitemap.js, robots.js, manifest.js (PWA manifest), opengraph-image.jsx (ImageResponse from next/og).
- next/font/local with preload and display: 'block'; next/image with remotePatterns for GitHub avatars; next/script with beforeInteractive for the theme.
- Hooks: useState, useEffect (with a cancellation flag to avoid stale updates), useLayoutEffect (paint before first frame), useRef.
- createPortal for the contributions modal, with a keydown listener to close on Escape.
- Composition: a shared layout kit (ResultHead, ResultSection, Bento, StatCell, RowList, Row, PersonCell).

Browser APIs

- localStorage: API cache, search history, personal token, theme preference.
- fetch with Authorization Bearer header and the GitHub Accept header.
- Canvas 2D (fillText, globalAlpha, setTransform for devicePixelRatio), requestAnimationFrame / cancelAnimationFrame.
- matchMedia('(prefers-reduced-motion: reduce)') to skip the animation for accessibility.
- Blob + URL.createObjectURL for file downloads; navigator.clipboard.writeText for the copy button.
- document keydown events, getComputedStyle to read CSS variables.

Architecture & design patterns

- **Service objects / module pattern:** CacheService, HistoryService, ExportService.
- **Cache-aside:** check the cache, otherwise fetch and store.
- **Progressive enhancement / hybrid rendering:** light server fetch for SEO and first paint, heavy data on the client.
- **Separation of concerns:** content (src/content) vs. presentation (components) vs. logic (lib).
- **Error mapping:** HTTP codes translated to domain errors with retry and token-entry recovery.
- **Strategy by input type:** analyzeQuery branches into repo / org / user pipelines.
- **Theming:** CSS custom properties with light/dark via a data-theme attribute set before hydration (no flash).
- **SEO:** JSON-LD (WebApplication and WebSite with SearchAction), Open Graph, Twitter cards, canonical URLs, curated sitemap.
- **Security & privacy:** the token is never sent to the server; password-manager autofill is disabled on the token input; poweredByHeader is off.